

Flex Server – How It Works with AdBuster

Flex Server is a lightweight local service that enables AdBuster to control your TV through Broadlink devices. It acts as a secure communication bridge between the AdBuster application and your Broadlink IR blaster, ensuring fast and fully local volume-automation.

What Flex Server Does

Flex Server manages all communication with your Broadlink device. Because Broadlink uses a proprietary encrypted LAN protocol, Windows applications cannot interact with it directly. Flex Server solves this by translating AdBuster's internal actions into the correct Broadlink IR commands.

AdBuster does not switch channels or perform advanced TV control. Its purpose is focused and intelligent: **detect sudden volume spikes, loud commercials, quiet dialogue, or abnormal audio changes — and automatically adjust the TV volume up or down.**

When an event is detected, AdBuster sends a minimal instruction to Flex Server (e.g., “volume down” or “volume up”), and the server delivers the corresponding IR signal to your TV.

ML/AI Detection and CEPA Logic

AdBuster's detection engine is powered by a combination of **machine learning**, **psychoacoustic modeling**, and **CEPA Logic** — a custom decision-making layer designed to mimic human-like reactions.

Machine Learning (ML/AI)

AdBuster continuously learns from audio patterns and adapts to:

- different TV models
- room acoustics
- user habits
- time-of-day behavior
- typical loudness of specific channels or programs

Over time, the system becomes more accurate at distinguishing:

- loud commercials
- artificially boosted audio
- quiet dialogue scenes
- natural vs. abnormal volume changes

This learning process improves responsiveness and reduces false triggers.

CEPA Logic

CEPA (Context-Enhanced Pattern Analysis) is the intelligence layer that interprets ML output and makes real-time decisions. It evaluates:

- context of the audio event
- intensity and duration
- historical patterns
- user-specific behavior
- time of day (e.g., quieter evenings, louder daytime)

The result is a reaction that feels **natural, human-like, and situationally aware** — not just “mute everything”, but **smart, adaptive volume control**.

How the System Works Together

1. **AdBuster monitors the audio stream** using ML/AI and psychoacoustic models.
2. **CEPA Logic interprets the event**, deciding whether to lower or raise the volume.
3. **AdBuster sends a simple command** (“volume down” or “volume up”) to Flex Server.
4. **Flex Server converts the instruction** into a Broadlink IR signal.
5. **The Broadlink device sends the IR command** to your TV, adjusting the volume instantly.

All communication remains **fully local**, with no cloud services involved.

Why This Architecture Matters

This modular design ensures:

- **maximum stability** — AdBuster focuses on detection, not hardware protocols
- **broad compatibility** with different Broadlink models
- **fast, reliable IR transmission**
- **continuous learning and adaptation** through ML/AI
- **human-like decision-making** via CEPA Logic
- **dynamic volume balancing** (both lowering and raising)
- **complete privacy**, as everything runs inside your home network

In Short

Flex Server is the local engine that transforms AdBuster’s intelligent audio-detection system into real-world, adaptive volume control — fast, human-like, and fully private.